

Midterm

gabriela davidson

2026-05-11

Set Up

```
# load packages
library(tidyverse)
library(janitor)
library(rstatix)
library(gt)

# read in halibut data
halibut <- read_csv("halibut.csv")

# read in bee pathogens data
bee_pathogens <- read_csv("bee-pathogens.csv")
```

Problems

Problem 1. Understanding, critiquing, and adding to written communication

1a. Test type

The analysis compared two groups (Northern vs Southern California) on a continuous variable (commercial landings in lbs) across time. There are two options for test type. The first is a parametric two-sample t-test (Welch's t-test). This would make sense if the halibut data are normally distributed. The Welch's t-test compares the means of two independent groups. The other test you could do is the Mann-Whitney U test which is non-parametric. This would make sense if the data is skewed or has outliers. This test compares the distributions of distributions/medians of two different groups. The similarities are that both tests compare

two independent groups on one continuous response variable and they both produce a p-value that is used to make a decision about a null hypothesis. The differences are that the t-test assumes normality and compares means whereas the Mann-Whitney test makes no normality assumption and compares medians.

1b. Writing new statements

T-Test statement: Across years (1969-2019), there was no significant difference in mean commercial halibut landings (lbs) between Northern (mean = 390,643 lbs, n = 51) and Southern California (mean = 350,715 lbs, n = 51) ($t(98.70) = 1.05$, $p = 0.30$).

Mann-Whitney Statement: Across years (1969-2019), there was no significant difference in median commercial halibut landings (lbs) between Northern (median = 387,626 lbs, n = 51) and Southern California (median = 329,274 lbs, n = 51) ($W = 1438$, $p = 0.36$).

1c. New figures

```
# creating a summary object with all necessary statistics
halibut_summary <- halibut |>
  # grouping by region to calculate stats separately for each group
  group_by(ca_region) |>
  summarize(
    # calculating mean landings for t-test figure
    mean_landings = mean(landings_lb),
    # calculating sample size
    n = length(landings_lb),
    # calculating degrees of freedom
    df = n - 1,
    # calculating standard deviation
    sd = sd(landings_lb),
    # calculating standard error
    se = sd / sqrt(n),
    # finding t-value for 95% confidence interval
    tval = qt(p = 0.05/2, df = df, lower.tail = FALSE),
    # calculating margin of error
    margin = tval * se,
    # calculating lower bound of 95% confidence interval
    ci_lower = mean_landings - margin,
    # calculating upper bound of 95% confidence interval
    ci_higher = mean_landings + margin,
    # calculating median landings for Mann-Whitney figure
```

```

    median_landings = median(landings_lb, na.rm = TRUE),
    # calculating first quartile for IQR
    q1 = quantile(landings_lb, 0.25, na.rm = TRUE),
    # calculating third quartile for IQR
    q3 = quantile(landings_lb, 0.75, na.rm = TRUE)
  ) |>
  # ungrouping after summarizing
  ungroup()

# running independent two-sample t-test comparing landings between regions
halibut_ttest <- t.test(
  landings_lb ~ ca_region, # landings as response, region as predictor
  data = halibut)          # using halibut data frame

# displaying t-test results
halibut_ttest

```

Welch Two Sample t-test

```

data:  landings_lb by ca_region
t = 1.0479, df = 98.695, p-value = 0.2972
alternative hypothesis: true difference in means between group Northern and group Southern is
95 percent confidence interval:
 -35675.85 115531.26
sample estimates:
mean in group Northern mean in group Southern
      390643.0           350715.3

```

```

# running Mann-Whitney U test as non-parametric alternative
halibut_mwu <- wilcox.test(
  landings_lb ~ ca_region, # landings as response, region as predictor
  data = halibut)          # using halibut data frame

# displaying Mann-Whitney U test results
halibut_mwu

```

Wilcoxon rank sum test with continuity correction

```

data:  landings_lb by ca_region

```

W = 1438, p-value = 0.3592

alternative hypothesis: true location shift is not equal to 0

Figure 1:

```
# --- Figure 1: T-test figure (means + 95% CI) ---

# creating ggplot base layer with region on x-axis and landings on y-axis
ggplot(data = halibut,
       mapping = aes(x = ca_region,
                     y = landings_lb,
                     color = ca_region)) +
# first layer: underlying data as small jittered points
# alpha makes them transparent to distinguish from summary stats
geom_jitter(width = 0.2,
            height = 0,
            alpha = 0.4,
            size = 1.5,
            shape = 16) +
# second layer: 95% confidence interval error bars from summary object
geom_errorbar(data = halibut_summary,
              mapping = aes(x = ca_region,
                            ymin = ci_lower,
                            ymax = ci_higher),
              inherit.aes = FALSE, # using summary data not halibut data
              width = 0.1,
              linewidth = 0.8) +
# third layer: mean as large filled point to distinguish from raw data
geom_point(data = halibut_summary,
           mapping = aes(x = ca_region,
                         y = mean_landings),
           size = 5,
           shape = 19) +
# custom colors different from ggplot defaults
scale_color_manual(values = c("Northern" = "blue",
                              "Southern" = "orange")) +
# clean axis labels with units
labs(x = "Geographic Region of California",
     y = "Commercial Halibut Landings (lbs)") +
# white panel background
theme_classic() +
```

```
# removing legend and grid lines
theme(legend.position = "none",
      panel.grid = element_blank(),
      panel.background = element_rect(fill = "white"))
```

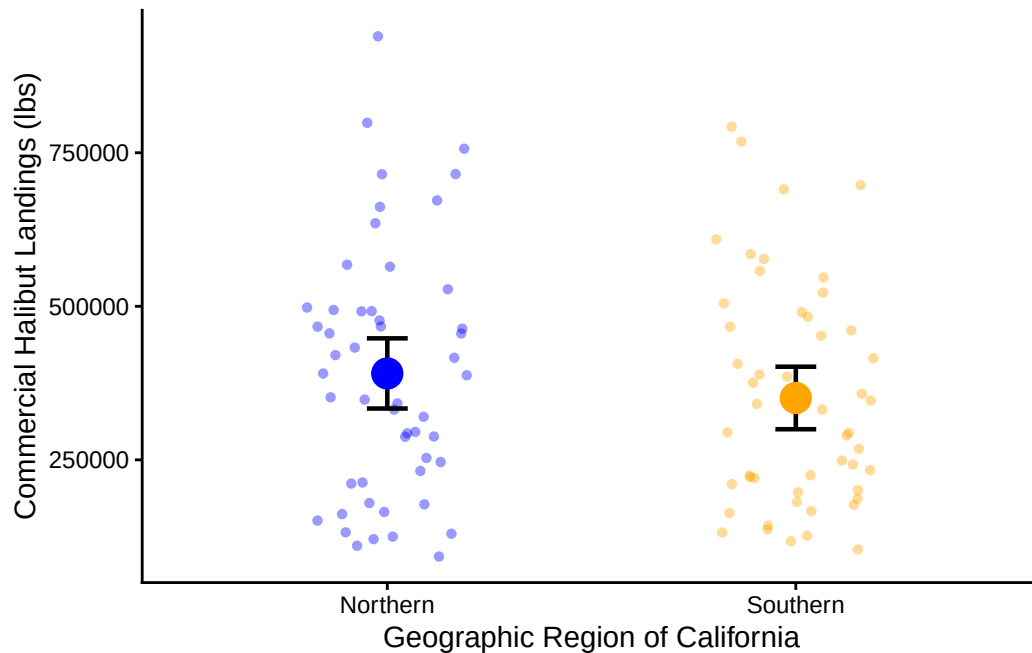


Figure 2:

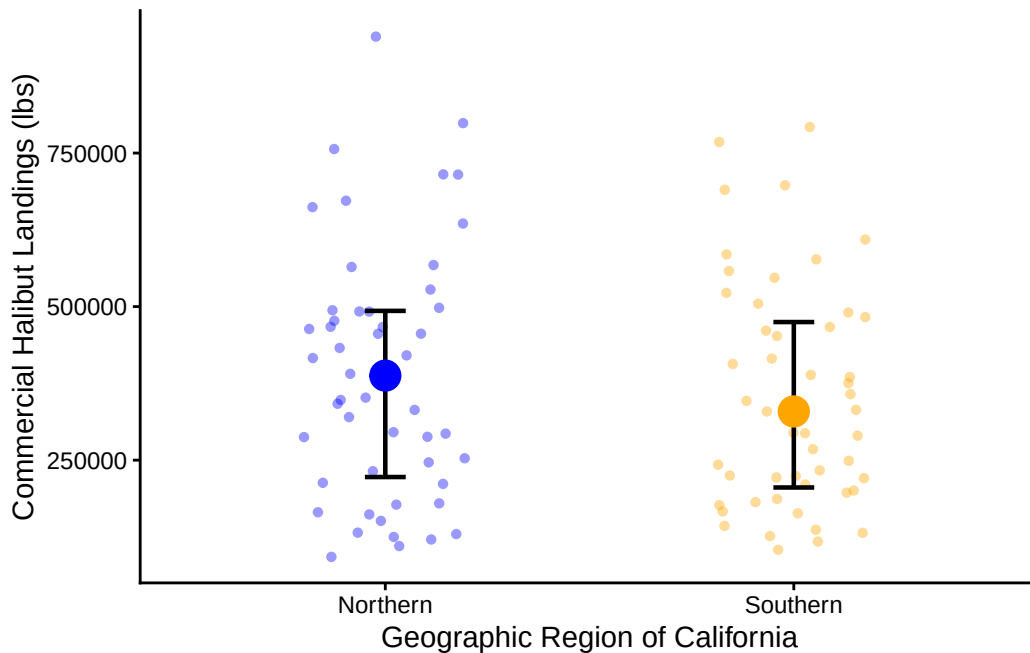
```
# --- Figure 2: Mann-Whitney figure (medians + IQR) ---

# creating ggplot base layer with region on x-axis and landings on y-axis
ggplot(data = halibut,
       mapping = aes(x = ca_region,
                     y = landings_lb,
                     color = ca_region)) +
# first layer: underlying data as small jittered points
# smaller size and lower alpha distinguishes them from median points
geom_jitter(width = 0.2,
            height = 0,
            alpha = 0.4,
            size = 1.5,
```

```

    shape = 16) +
# second layer: IQR (Q1 to Q3) as error bars from summary object
geom_errorbar(data = halibut_summary,
  mapping = aes(x = ca_region,
    ymin = q1,
    ymax = q3),
  inherit.aes = FALSE, # using summary data not halibut data
  width = 0.1,
  linewidth = 0.8) +
# third layer: median as large filled point to distinguish from raw data
geom_point(data = halibut_summary,
  mapping = aes(x = ca_region,
    y = median_landings),
  size = 5,
  shape = 19) +
# custom colors different from ggplot defaults
scale_color_manual(values = c("Northern" = "blue",
  "Southern" = "orange")) +
# clean axis labels with units
labs(x = "Geographic Region of California",
  y = "Commercial Halibut Landings (lbs)") +
# white panel background
theme_classic() +
# removing legend and grid lines
theme(legend.position = "none",
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "white"))

```



1d. Caption Writing

Figure 1. Northern and Southern California commercial halibut landings did not differ across years (1969-2019). Points show mean commercial halibut (*Paralichthys californicus*) landings (lbs) for Northern (blue) and Southern (orange) California. Error bars show 95% confidence intervals. Underlying annual values are shown as smaller, transparent jittered points. California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Figure 2. Northern and Southern California commercial halibut landings did not differ across years (1969-2019). Large points show median commercial halibut (*Paralichthys californicus*) landings (lbs) for Northern (blue) and Southern (orange) California. Error bars show the interquartile range (Q1-Q3). Underlying annual values are shown as smaller, transparent jittered points. California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 2. Data visualization

2a. Plan your figure

- x-axis (name the specific column): `year`
- y-axis (name the specific column): `landings_lb`

- geometry function (hint: there is only one): `geom_line()`
- colors and linetypes (name the specific column): `ca_region`
- relabelling the x- and y-axes (name the specific function): `labs()`
- using custom colors (name the specific function): `scale_color_manual()`
- theme arguments and the `element_` for:
 - positioning the x-axis text at a 45° angle: `axis.text.x = element_text(angle = 45, hjust = 1)`
 - putting the legend in the top right of the plot: `legend.position = c(0.9, 0.9)`
 - removing the panel background: `panel.background = element_blank()`
 - removing the panel grid: `panel.grid = element_blank()`
 - removing the axis ticks: `axis.ticks = element_blank()`
 - removing the legend title: `legend.title = element_blank()`

2b. Code your figure

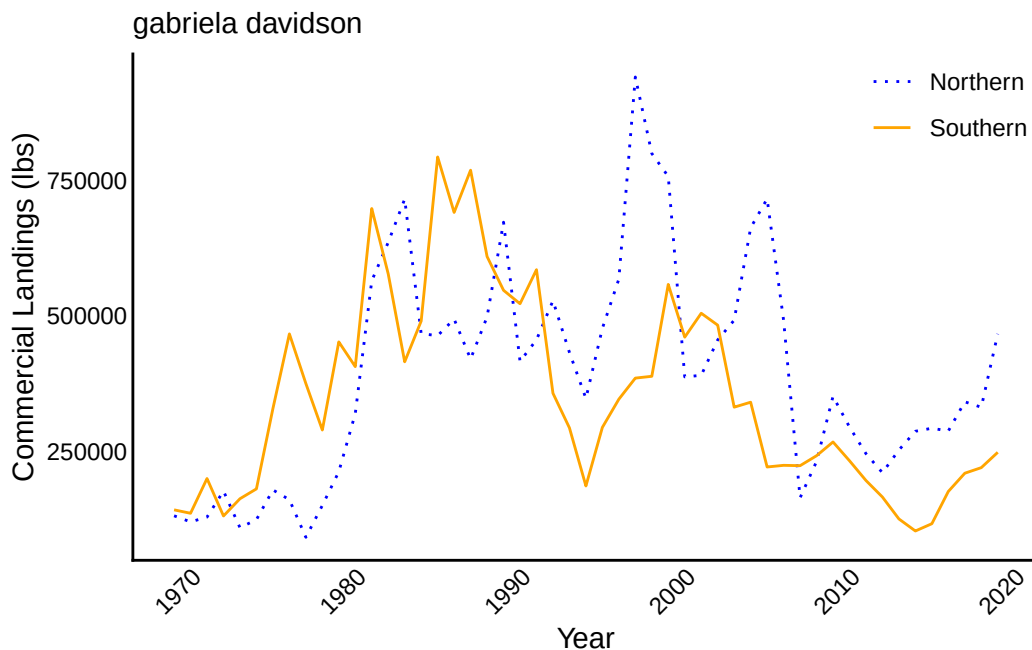
```
# creating ggplot base layer with year on x-axis and landings on y-axis
ggplot(data = halibut,
       mapping = aes(x = year,
                     y = landings_lb,
                     color = ca_region,
                     linetype = ca_region)) +
# creating line geometry for time series
geom_line() +
# setting custom colors for each region to differ from ggplot defaults
scale_color_manual(values = c(
  "Northern" = "blue",
  "Southern" = "orange")) +
# setting custom linetypes for each region to match reference figure
scale_linetype_manual(values = c(
  "Northern" = "dotted",
  "Southern" = "solid")) +
# relabeling axes and adding name as subtitle
labs(x = "Year",
     y = "Commercial Landings (lbs)",
     subtitle = "gabriela davidson") +
# applying classic theme as base
theme_classic() +
# customizing individual theme elements
theme(
  axis.text.x = element_text(angle = 45, hjust = 1),
```



```

legend.position = c(0.9, 0.9),
panel.background = element_blank(),
panel.grid = element_blank(),
axis.ticks = element_blank(),
legend.title = element_blank()

```



2c. Write a Caption

Figure 3. Commercial halibut landings fluctuated over time in both Northern and Southern California from 1969-2019. Lines show annual commercial halibut (*Paralichthys californicus*) landings (lbs) for Northern (blue, dotted line) and Southern (orange, solid line) California. California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 3. Reproducing an analysis

3a. Variables

Response variable: Black Queen Cell Virus (BQCV) load is a continuous numerical variable measuring the amount of virus detected in bumble bees. Predictor variable: Habitat type is a categorical variable with three groups: valley, edge, and forest. How measured: BQCV load was

measured using quantitative PCR (qPCR) performed on pooled abdomens of three *Bombus impatiens* workers per sample. RNA was extracted and reverse transcribed to cDNA, then amplified using BQCV-specific primers in a SybrGreen-based reaction. CT values above 35 were considered “no detection” and assigned a value of 35, and results were normalized using 2 delta-delta Ct to obtain log2 fold differences from no detection. This process is described in the Pathogen Quantification subsection of the Methods section of Gratton et al. 2022 (paragraphs 1-3).

3b. Test type

The researchers used a Kruskal-Wallis test to assess the effect of habitat type on BQCV loads, followed by a Dunn post-hoc test with a Bonferroni correction to adjust for multiple comparisons. This was described in the Statistical Analysis subsection of the Methods section of Gratton et al. 2022 (paragraph 1).

3c. Hypotheses

H0: The median BQCV load is the same across all habitat types (valley = edge = forest).

Ha: At least one habitat type has a different median BQCV load from the others.

3d. Cleaning code

```
# creating cleaned data object from raw bee_pathogens data
bee_pathogens_clean <- bee_pathogens |>
  clean_names() |> # standardizing all column names to snake_case
  select(site_type_x, bqcv) |> # selecting only the site type and BQCV columns
  rename(habitat = site_type_x) # renaming site_type_x to habitat

# displaying 10 random rows from bee_pathogens_clean to verify cleaning worked
slice_sample(bee_pathogens_clean, n = 10)
```

```
# A tibble: 10 x 2
  habitat    bqcv
  <chr>    <dbl>
1 edge      8.15
2 valley    9.42
3 valley    NA
4 valley    7.79
5 forest     3.98
```

```

6 forest    1.000
7 forest    5.99
8 edge      3.50
9 valley    8.45
10 edge     8.72

```

3e. Summarizing and table displays

```

# creating summary object from cleaned bee pathogens data
bee_pathogens_summary <- bee_pathogens_clean |>
  group_by(habitat) |> # grouping by habitat type
  summarize(
    median_bqcv = round(median(bqcv, na.rm = TRUE), 1),
    # calculating median BQCV load rounded to 1 decimal place
    n = n() # calculating sample size for each habitat group
  ) |>
  ungroup() # removing grouping structure after summarizing

```

```

# displaying summary as polished gt table
bee_pathogens_summary |>
  gt() |>
  # renaming columns to be polished with no underscores and sentence case
  cols_label(
    habitat = "Habitat",
    median_bqcv = "Median BQCV load",
    n = "Sample size"
  ) |>
  # new function to position table correctly in rendered PDF
  tab_options(
    latex.tbl.pos = "h", # h for render the table "here"
    latex.use_longtable = TRUE, # setting PDF rendering option
    table.width = px(500) # making the table wider
  )

```

Habitat	Median BQCV load	Sample size
edge	6.3	31
forest	5.0	26
valley	7.6	30

3f. Do a test

```
# running Kruskal-Wallis test to assess whether BQCV load differs across habitat types
kruskal.test(
  bqcv ~ habitat, # BQCV load as response variable, habitat type as predictor
  data = bee_pathogens_clean # using cleaned bee pathogens data frame
)
```

Kruskal-Wallis rank sum test

data: bqcv by habitat

Kruskal-Wallis chi-squared = 9.8219, df = 2, p-value = 0.007365

3g. Do a post hoc test

```
# running Dunn post-hoc test to identify which specific habitat pairs differ significantly
dunn_test(
  bee_pathogens_clean, # using cleaned bee pathogens data frame
  bqcv ~ habitat, # BQCV load as response variable, habitat type as predictor
  p.adjust.method = "bonferroni"
  # applying Bonferroni correction to adjust p-values for multiple comparisons
)
```

A tibble: 3 x 9

	.y.	group1	group2	n1	n2	statistic	p	p.adj	p.adj.signif
*	<chr>	<chr>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<chr>
1	bqcv	edge	forest	27	24	-1.83	0.0679	0.204	ns
2	bqcv	edge	valley	27	29	1.31	0.189	0.567	ns
3	bqcv	forest	valley	24	29	3.13	0.00175	0.00525	**

3h. Calculate an effect size

```
# calculating eta-squared effect size for the Kruskal-Wallis test
kruskal_effsize(
  bee_pathogens_clean, # using cleaned bee pathogens data frame
  bqcv ~ habitat # BQCV load as response variable, habitat type as predictor
)
```

```
# A tibble: 1 x 5
  .y.      n effsize method  magnitude
* <chr> <int>   <dbl> <chr>   <ord>
1 bqcv    87  0.0931 eta2[H] moderate
```

3i. Written communication

Habitat type had a significant effect on BQCV loads (Kruskal-Wallis: $H(2) = 9.82$, $p = 0.007$, $\alpha = 0.05$, $n = 87$). The effect size was moderate ($\eta^2 = 0.09$). Valley habitats had significantly higher BQCV loads (median = 7.6, $n = 30$) than forest habitats (median = 5.0, $n = 26$; $z = 3.13$, $p_{\text{adj}} = 0.005$), but edge habitats (median = 6.3, $n = 31$) did not differ significantly from either habitat type.